

4.5 Manipulating Power Series Worksheet

1. Answer each of the following questions

a) Find a series that sums to equal $\ln(1.5)$.

We know that $\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} = \ln(x+1)$ for x values in $(-1, 1]$.

If we let $x = 0.5$ then we can say that $\sum_{n=0}^{\infty} \frac{(-1)^n 0.5^{n+1}}{n+1} = \ln(1.5)$.

b) Why can't you easily repeat the process from part (a) to find a series that sums to be $\ln(10)$?

We couldn't use $x = 9$ and repeat the process from above because 9 is not in the interval of convergence for the series.

c) Find a power series representation for $x \ln(x+1)$.

We know that $\sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} = \ln(x+1)$ for x values in $(-1, 1]$.

Therefore, $x \cdot \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+1}}{n+1} = x \cdot \ln(x+1) \rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+2}}{n+1} = x \ln(x+1)$ for x values in $(-1, 1]$.

d) Find a series that sums to $\ln\left(\sqrt[3]{\frac{4}{3}}\right)$.

We may notice that $\ln\left(\sqrt[3]{\frac{4}{3}}\right) = \frac{1}{3} \ln\left(\frac{4}{3}\right) = \frac{1}{3} \ln\left(\frac{1}{3} + 1\right)$. This resembles the series that we found in

part (c) with $x = \frac{1}{3}$. Since this value is inside the interval of convergence of the series we can

say that $\ln\left(\sqrt[3]{\frac{4}{3}}\right) = \sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{1}{3}\right)^{n+2}}{n+1}$.

2. What does the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ converge to?

We recall that $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{2n+1} = \tan^{-1}(x)$ for x values in $[-1, 1]$. The given series resembles this series

with $x = 1$. Therefore we can say that $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} = \tan^{-1}(1) = \frac{\pi}{4}$.

3. Find a power series representation for $\sin(x^2)$.

We know that $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$ for all x values.

Therefore we can say that $\sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}$ for all x values.

4. Consider the integral $\int_0^1 \cos(x^3) dx$.

a) Find a series which represents this integral.

We know that $\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ for all x values.

Therefore, we can say that $\cos(x^3) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^3)^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{6n}}{(2n)!}$.

Then we can calculate that

$$\int_0^1 \cos(x^3) dx = \sum_{n=0}^{\infty} \frac{(-1)^n x^{6n+1}}{(2n)!(6n+1)} \bigg|_0^1 = \sum_{n=0}^{\infty} \frac{(-1)^n (1)^{6n+1}}{(2n)!(6n+1)} - \sum_{n=0}^{\infty} \frac{(-1)^n (0)^{6n+1}}{(2n)!(6n+1)} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!(6n+1)}$$

b) Approximate the value of this integral by calculating the 3rd partial sum of the series.

$$\int_0^1 \cos(x^3) dx \approx s_2 = 1 - \frac{1}{14} + \frac{1}{312} = \frac{2035}{2184} \approx 0.9318$$

- c) Use a theorem from 3.6 to determine what bound you can place on the error of this approximation.

Since the given series is an alternating series with $a_{n+1} \leq a_n$ for all n and $\lim_{n \rightarrow \infty} a_n = 0$, then

$$|s_n - S| \leq a_{n+1}.$$

Therefore we can say that $|s_2 - S| \leq a_3$ which means that $\left| \frac{2035}{2184} - S \right| \leq \frac{1}{13,680}$.

So the value of $\frac{2035}{2184}$ is at most $\frac{1}{13,680}$ away from the exact value of the integral.

5. Consider the function $f(x) = \ln(1+x)$.

- a) Determine a power series representation for $f(x)$.

We already completed this task in class (see 6.3 example 5), so we know that

$$f(x) = \ln(1+x) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{n+1}}{n+1} \right) \text{ for } x \text{ in } (-1, 1].$$

- b) Use the power series from part(a) to help determine the value of the following series

$$\sum_{k=3}^{\infty} \frac{(-1)^{k+1}}{k} = \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \dots$$

If we compare this series to one we created in part (a) we realize that they are pretty similar if we let $x=1$.

$$\ln(1+1) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n (1)^{n+1}}{n+1} \right) = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \dots$$

So we can write that :

$$\ln(2) = 1 - \frac{1}{2} + \sum_{k=3}^{\infty} \frac{(-1)^{k+1}}{k} \Leftrightarrow \ln(2) - \frac{1}{2} = \sum_{k=3}^{\infty} \frac{(-1)^{k+1}}{k}$$

Therefore the series converges to $\ln(2) - \frac{1}{2}$.

6. Consider the function $f(x) = \frac{1}{1-x}$.

a) Determine a power series representation for $f(x)$.

We recognize this function as the sum of a geometric series with $a=1$ and $r=x$. Therefore

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \text{ provided that } |x| < 1 \Leftrightarrow -1 < x < 1.$$

b) Use the power series from part(a) to help determine the value of the following series

$$\sum_{k=1}^{\infty} \frac{k+1}{2^k} = \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \frac{5}{2^4} + \dots$$

(Hint: It will be helpful to manipulate the power series from part(a) using techniques from 6.5)

From part(a) we have $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$. We will have to play around with this given information to try and form the series we are trying to find the sum of.

If we differentiate the series we obtain $\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}$.

If we then replace x by $\frac{1}{x}$ we obtain $\sum_{k=1}^{\infty} k \left(\frac{1}{x}\right)^{k-1} = \frac{1}{\left(1-\frac{1}{x}\right)^2}$.

Through a bit of rewriting we see that $\sum_{k=1}^{\infty} k \left(\frac{1}{x}\right)^{k-1} = \sum_{k=1}^{\infty} \frac{k}{x^{k-1}} = \frac{1}{\left(1-\frac{1}{x}\right)^2}$.

If we let $x=2$ then we see that $\sum_{k=1}^{\infty} \frac{k}{(2)^{k-1}} = \frac{1}{1} + \frac{2}{2} + \frac{3}{2^2} + \frac{4}{2^3} + \frac{5}{2^4} + \dots$

Therefore we can write that $1 + \sum_{k=1}^{\infty} \frac{k+1}{2^k} = \sum_{k=1}^{\infty} \frac{k}{(2)^{k-1}} = \frac{1}{\left(1-\frac{1}{2}\right)^2}$ which implies that $\sum_{k=1}^{\infty} \frac{k+1}{2^k} = \frac{1}{\left(1-\frac{1}{2}\right)^2} - 1$

and when simplified this means that $\sum_{k=1}^{\infty} \frac{k+1}{2^k} = 3$

7. For x in the interval $(-1,1)$, what is the sum of the following series? (Hint: It will be helpful to manipulate the power series for $\frac{1}{1+x}$ using techniques from 6.5)

$$\sum_{n=1}^{\infty} (-1)^{n+1} nx^n = x - 2x^2 + 3x^3 - 4x^4 + 5x^5 - 6x^6 + \dots$$

We first note that $\frac{1}{1+x} = \frac{1}{1-(-x)}$. We know that this is the sum of the geometric series with $a=1$

and $r=-x$ provided that $|-x| < 1 \Leftrightarrow -1 < x < 1$. Therefore we have $\frac{1}{1+x} = \sum_{n=0}^{\infty} ((-x)^n) = \sum_{n=0}^{\infty} ((-1)^n x^n)$.

We will have to play around with this information to try and form the series we are trying to find the sum of.

If we differentiate the series we obtain $\sum_{n=1}^{\infty} (-1)^n nx^{n-1} = \frac{-1}{(1+x)^2}$.

If we multiply the series by $-x$ we obtain $\sum_{n=1}^{\infty} (-1)^{n+1} nx^n = \frac{x}{(1+x)^2}$.

This is exactly the series we were trying to find! So we have found the sum of this series to be $\frac{x}{(1+x)^2}$

(provided that $-1 < x < 1$).

8. Consider the function $f(x) = \tan^{-1}(x^2)$.

a) Determine a powers series representation for this function and state its interval of convergence.

In class we established that $\tan^{-1}(x) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{2n+1}}{2n+1} \right)$ for $-1 \leq x \leq 1$. If we substitute x^2 for x we

obtain $\tan^{-1}(x^2) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n (x^2)^{2n+1}}{2n+1} \right) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{4n+2}}{2n+1} \right)$ for $-1 \leq x \leq 1$.

b) Use part(a) to help show that $\int_0^{1/10} \tan^{-1}(x^2) dx \approx \frac{1}{3000}$.

Since the interval $\left[0, \frac{1}{10}\right]$ is inside the interval of convergence we can perform this integration. We have that

$$\begin{aligned} \int_0^{1/10} \tan^{-1}(x^2) dx &= \int_0^{1/10} \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{4n+2}}{2n+1} \right) dx = \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{4n+3}}{(2n+1)(4n+3)} \right) \bigg|_0^{1/10} \\ &= \sum_{n=0}^{\infty} \left(\frac{(-1)^n \left(\frac{1}{10} \right)^{4n+3}}{(2n+1)(4n+3)} \right) - \sum_{n=0}^{\infty} \left(\frac{(-1)^n (0)^{4n+3}}{(2n+1)(4n+3)} \right) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n}{(2n+1)(4n+3)10^{4n+3}} \right) - 0 \\ &= \frac{1}{3 \cdot 10^3} - \frac{1}{3 \cdot 7 \cdot 10^7} + \frac{1}{5 \cdot 11 \cdot 10^{11}} - \frac{1}{7 \cdot 15 \cdot 10^{15}} + \dots \\ &\approx \frac{1}{3 \cdot 10^3} = \frac{1}{3000} \end{aligned}$$

c) Use a theorem from 3.6 to determine what bound you can place on the error of this approximation.

Since the given series is an alternating series with $a_{n+1} \leq a_n$ for all n and $\lim_{n \rightarrow \infty} a_n = 0$, then

$$|s_n - S| \leq a_{n+1}.$$

Therefore we can say that $|s_0 - S| \leq a_1$ which means that $\left| \frac{1}{3000} - S \right| \leq \frac{1}{3 \cdot 7 \cdot 10^7} = \frac{1}{210,000,000}$.

So the value of $\frac{1}{3000}$ is at most $\frac{1}{210,000,000}$ away from the exact value of the integral.

Note: This is *really* impressive we have roughly 8 decimal places of accuracy by only looking at the first term of the series! Without this technique, we currently have no ability to evaluate such an integral!!!